

ResumeAI - Project Architecture & Viva Guide

Intelligent Multi-Stage Resume Analyzer, Deterministic ATS Scorer & Semantic Job Matcher

1. Project Summary & Purpose

ResumeAI is a full-stack, AI-powered career mentorship and resume intelligence application. It bridges the gap between student resumes and modern hiring expectations by combining **deterministic ATS compliance scoring**, **bidirectional skill taxonomy normalization**, **3-tier transferable skill matching**, **role-aware learning roadmaps**, and a **30+ question interview preparation coach**.

2. Important Folders & Core Modules

Directory / File	Role & Architectural Purpose
run.bat	1-click automatic startup script: checks Python, creates virtual environment, installs dependencies, and launches browser.
backend/main.py	FastAPI server entry point, CORS middleware, API route registration, and static frontend asset router.
backend/nlp/	Core NLP parsing engine: Multi-model OCR fallback (parser.py), heading/contact extraction (section_extractor.py), 200+ skill taxonomy (skill_taxonomy.py), and SBERT vector matching (semantic_matcher.py).
backend/ai/gemini_client.py	Multi-model Google Gemini client: generates role-specific learning roadmaps, 30+ categorized interview questions with STAR answers, and high-impact bullet rewrites.
backend/services/scoring_service.py	Computes 3 distinct scores: Resume Health (0-100), ATS Compatibility (0-100), and Job Match %, with itemized verification evidence.
frontend/	Modern, responsive web interface built with HTML5, CSS3, Vanilla JavaScript, and Chart.js (Radar, Donut, and Bar charts).
resume_ai.db	Local SQLite database with auto-fallback from MySQL, storing user accounts, resume metadata, extracted skills, and historical analysis.

3. End-to-End Analysis Pipeline (5 Stages)

Stage 1: Multi-Engine Document Parsing: Local text extraction via pdfplumber and docx with automated multi-model Gemini OCR fallback for graphic Canva PDFs.

Stage 2: Robust Contact & Heading Detection: Extracts international phone numbers, email, LinkedIn, GitHub, Name, and standard section boundaries.

Stage 3: Canonical Skill Normalization: Normalizes variations (e.g. HTML5 to HTML, JS to JavaScript, ReactJS to React) across all document sections.

Stage 4: 3-Tier Job Matching: Categorizes requirements into Matched, Partially Matched (Transferable), and Missing with SentenceTransformer vector similarity.

Stage 5: Generative Mentorship & Interview Prep: Generates role-aware weekly roadmaps (with YouTube/doc links), non-hallucinatory bullet rewrites, and 30+ categorized interview questions.

4. Frequently Asked Viva / Project Evaluation Questions

Q1: How does your ATS verification differ from standard keyword counters?

A: Traditional ATS checkers rely on literal substring matches which fail on aliases and graphic PDFs. ResumeAI utilizes multi-model OCR fallback, canonical taxonomy normalization, transferable skill mapping (e.g. MySQL to PostgreSQL), and SBERT dense vector embeddings.

Q2: How is the overall Job Match percentage calculated?

A: The Job Match percentage is a weighted formula combining: 3-Tier Skill Overlap (55%), Semantic SBERT Cosine Similarity (30%), and Experience Relevance Context (15%).

Q3: How do you guarantee the AI does not invent fake metrics in resume rewrites?

A: The Gemini prompt strictly forbids hallucinating numbers, percentages, or companies. If a metric is absent in the user resume, it provides a metric placeholder suggestion.

Q4: What happens if MySQL is offline or not installed?

A: The backend database engine in backend/database.py includes an automatic resilient fallback that seamlessly switches to local SQLite (resume_ai.db) with zero downtime.